

Introduction to **Information Retrieval**

Text Classification

Standing queries

- The path from IR to text classification:
 - You have an information need to monitor, say:
 - **Unrest in the Niger delta region**
 - You want to rerun an appropriate query periodically to find new news items on this topic
 - You will be sent new documents that are found
 - I.e., it's not **ranking** but **classification** (relevant vs. not relevant)
- Such queries are called **standing queries**
 - Long used by “information professionals”
 - A modern mass instantiation is **Google Alerts**
- Standing queries are (hand-written) text classifiers

From: Google Alerts

Subject: Google Alert - stanford -neuro-linguistic nlp OR "Natural Language Processing" OR parser OR tagger OR ner OR "named entity" OR segmenter OR classifier OR dependencies OR "core nlp" OR corenlp OR phrasal

Date: May 7, 2012 8:54:53 PM PDT

To: Christopher Manning

Web

3 new results for stanford -neuro-linguistic nlp OR "Natural Language Processing" OR parser OR tagger OR ner OR "named entity" OR segmenter OR classifier OR dependencies OR "core nlp" OR corenlp OR phrasal

[Twitter / Stanford NLP Group: @Robertoross If you only n ...](#)

@Robertoross If you only need tokenization, java -mx2m edu.stanford.nlp.process.PTBTOKENIZER file.txt runs in 2MB on a whole file for me.... 9:41 PM Apr 28th ...

twitter.com/stanfordnlp/status/196459102770171905

[\[Java\] LexicalizedParser lp = LexicalizedParser.loadModel\("edu ...](#)

loadModel("edu/stanford/nlp/models/lexparser/englishPCFG.ser.gz");. String[] sent = { "This", "is", "an", "easy", "sentence", "." };. Tree parse = lp.apply(Arrays.

pastebin.com/az14R9nd

[More Problems with Statistical NLP || kuro5hin.org](#)

Tags: nlp, ai, coursera, stanford, nlp-class, cky, nltk, reinventing the wheel, ... Programming Assignment 6 for Stanford's nlp-class is to implement a CKY parser .

www.kuro5hin.org/story/2012/5/5/11011/68221

Tip: Use quotes ("like this") around a set of words in your query to match them exactly. [Learn more.](#)

[Delete](#) this alert.

[Create](#) another alert.

[Manage](#) your alerts.

Spam filtering

Another text classification task

From: "" <takworld@hotmail.com>

Subject: real estate is the only way... gem oalvgkay

Anyone can buy real estate with no money down

Stop paying rent TODAY !

There is no need to spend hundreds or even thousands for similar courses

I am 22 years old and I have already purchased 6 properties using the methods outlined in this truly INCREDIBLE ebook.

Change your life NOW !

=====

Click Below to order:

<http://www.wholesaledaily.com/sales/nmd.htm>

Categorization/Classification

- Given:

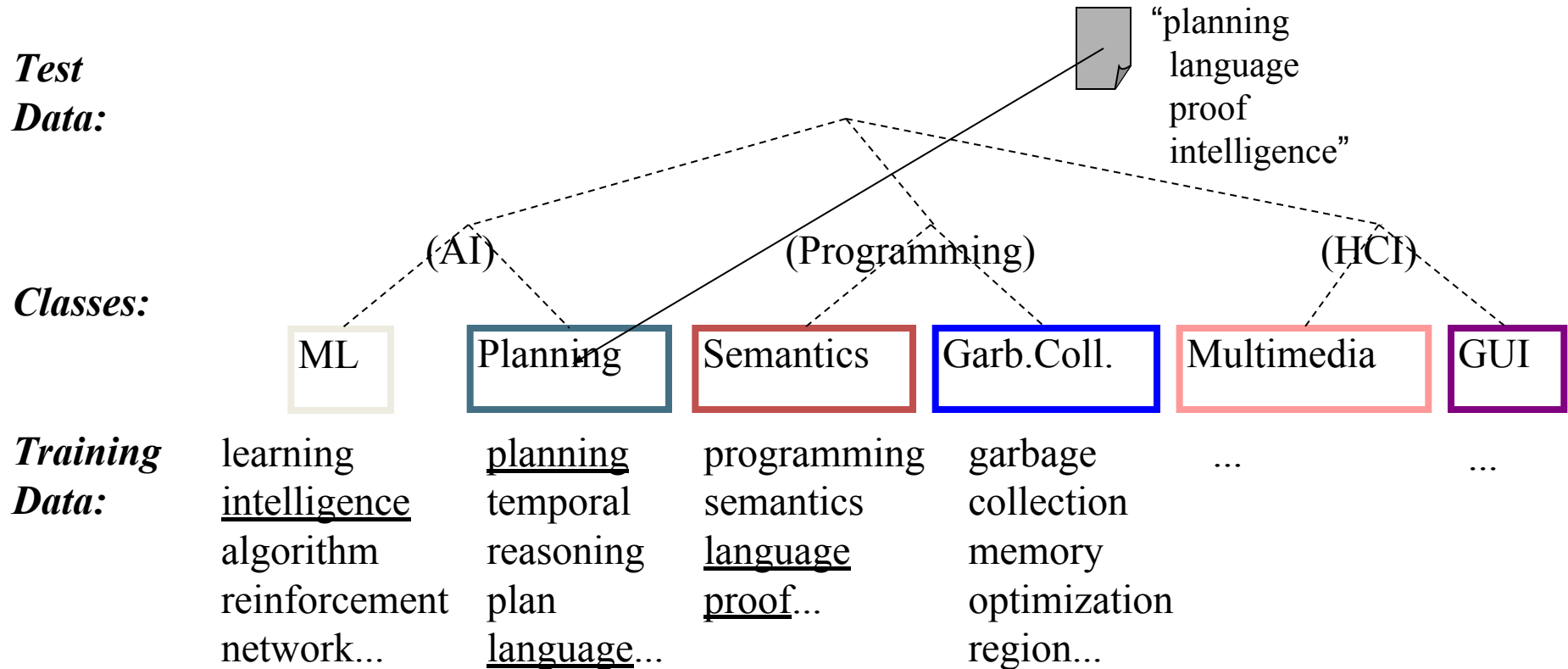
- A representation of a document d
 - Issue: how to represent text documents.
 - Usually some type of high-dimensional space – bag of words
- A fixed set of classes:

$$C = \{c_1, c_2, \dots, c_J\}$$

- Determine:

- The category of d : $\gamma(d) \in C$, where $\gamma(d)$ is a classification function
- We want to build classification functions (“classifiers”).

Document Classification



Classification Methods (1)

- Manual classification
 - Used by the original Yahoo! Directory
 - Looksmart, about.com, ODP, PubMed
 - Accurate when job is done by experts
 - Consistent when the problem size and team is small
 - Difficult and expensive to scale
 - Means we need automatic classification methods for big problems

Classification Methods (2)

- Hand-coded rule-based classifiers
 - One technique used by news agencies, intelligence agencies, etc.
 - Widely deployed in government and enterprise
 - Vendors provide “IDE” for writing such rules

Classification Methods (2)

- Hand-coded rule-based classifiers
 - Commercial systems have complex query languages
 - Accuracy can be high if a rule has been carefully refined over time by a subject expert
 - Building and maintaining these rules is expensive

A Verity topic

A complex classification rule: art

```

comment line      # Beginning of art topic definition
top-level topic   art ACCRUE

topic definition modifiers {
    /author = "fsmith"
    /date  = "30-Dec-01"
    /annotation = "Topic created
                    by fsmith"

subtopic topic    * 0.70 performing-arts ACCRUE
evidencetopic     ** 0.50 WORD
topic definition modifier /wordtext = ballet
evidencetopic     ** 0.50 STEM
topic definition modifier /wordtext = dance
evidencetopic     ** 0.50 WORD
topic definition modifier /wordtext = opera
evidencetopic     ** 0.30 WORD
topic definition modifier /wordtext = symphony
subtopic          * 0.70 visual-arts ACCRUE
                  ** 0.50 WORD
                  /wordtext = painting
                  ** 0.50 WORD
                  /wordtext = sculpture
subtopic          * 0.70 film ACCRUE
                  ** 0.50 STEM
                  /wordtext = film
subtopic          ** 0.50 motion-picture PHRASE
                  *** 1.00 WORD
                  /wordtext = motion
                  *** 1.00 WORD
                  /wordtext = picture
                  ** 0.50 STEM
                  /wordtext = movie
subtopic          * 0.50 video ACCRUE
                  ** 0.50 STEM
                  /wordtext = video
                  ** 0.50 STEM
                  /wordtext = vcr
                  # End of art topic

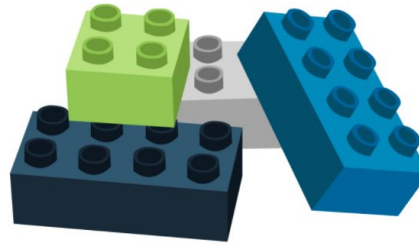
```

■ Note:

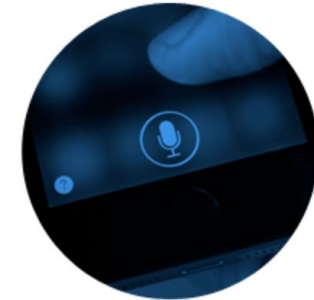
- maintenance issues (author, etc.)
- Hand-weighting of terms

[Verity was bought by Autonomy in 2005, which was bought by HP in 2011 – a mess; I think it no longer exists ...]

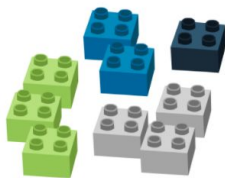
At the heart of eContext is its taxonomy of the commercial and social web. This taxonomy provides a consistent framework that groups conversations, behavior, and digital interactions into topical hierarchies up to 21 tiers, with clear connections between those topics. Clients push large amounts of text data from multiple sources through eContext, creating a single data set for measurement and analytics. With this holistic perspective, they can distill more comprehensive, meaningful insights, in less time.



Schedule a Demo



Enhancing Virtual Agents with Structured Knowledge



As clients feed information through our API, our custom pre-processing parses the text into linguistically appropriate units that eContext can work with efficiently.

Then eContext checks each unit against its 55 million positive and negative rules to determine the correct topic, or topics, that content exhibits. This happens hundreds of thousands of times per second, as we enrich the highest velocity data streams available in our industry — everything up to and including the full firehose of Tweets. Thanks to the rules that govern eContext's decision-making, and the 450,000 different concepts it recognizes, eContext is able to deliver deep, broad, and accurate results.



Download White Paper

Classification Methods (3): Supervised learning

- Given:
 - A document d
 - A fixed set of classes:
$$\mathcal{C} = \{c_1, c_2, \dots, c_J\}$$
 - A training set D of documents each with a label in $\mathcal{C}$
- Determine:
 - A learning method or algorithm which will enable us to learn a classifier γ
 - For a test document d , we assign it the class
$$\gamma(d) \in \mathcal{C}$$

Classification Methods (3)

- Supervised learning
 - Naive Bayes (simple, common) – see video, cs229
 - k-Nearest Neighbors (simple, powerful)
 - Support-vector machines (newer, generally more powerful)
 - Decision trees □ random forests □
gradient-boosted decision trees (e.g., xgboost)
 - ... plus many other methods
 - No free lunch: need hand-classified training data
 - But data can be built up by amateurs
- Many commercial systems use a mix of methods

Features

- Supervised learning classifiers can use any sort of feature
 - URL, email address, punctuation, capitalization, dictionaries, network features
- In the simplest bag of words view of documents
 - We use **only** word features
 - we use **all** of the words in the text (not a subset)

The bag of words representation

 γ (

I love this movie! It's sweet, but with satirical humor. The dialogue is great and the adventure scenes are fun.. It manages to be whimsical and romantic while laughing at the conventions of the fairy tale genre. I would recommend it to just about anyone. I've seen it several times, and I'm always happy to see it again whenever I have a friend who hasn't seen it yet.

 $) = c$

The bag of words representation

$\gamma($

great	2
love	2
recommend	1
laugh	1
happy	1
...	...

$) = c$

Naïve Bayes: See IIR 13 or cs124 lecture on Coursera or cs229

- Classify based on prior weight of class and conditional parameter for what each word says:

$$c_{NB} = \operatorname{argmax}_{c_j \in C} \left[\log P(c_j) + \sum_{i \in \text{positions}} \log P(x_i | c_j) \right]$$

- Training is done by counting and dividing:

$$P(c_j) \leftarrow \frac{N_{c_j}}{N} \quad P(x_k | c_j) \leftarrow \frac{T_{c_j x_k} + \alpha}{\sum_{x_i \in V} [T_{c_j x_i} + \alpha]}$$

- Don't forget to smooth

Naive Bayes is Not So Naive

- Very fast learning and testing (basically just count words)
- Low storage requirements
- Very good in domains with many equally important features
- More robust to irrelevant features than many learning methods

Irrelevant features cancel out without affecting results

Naive Bayes is Not So Naive

- More robust to concept drift (changing class definition over time)
- Naive Bayes won 1st and 2nd place in KDD-CUP 97 competition out of 16 systems

Goal: Financial services industry direct mail response prediction: Predict if the recipient of mail will actually respond to the advertisement – 750,000 records.

- A good dependable baseline for text classification (but not the best)!

Evaluating Categorization

- Evaluation must be done on test data that are independent of the training data
 - Sometimes use cross-validation (averaging results over multiple training and test splits of the overall data)
- Easy to get good performance on a test set that was available to the learner during training (e.g., just memorize the test set)